Compute Cell Scheduling Records

This document records how tasks were scheduled on small compute clusters. The workload and the machine failures are real records from a production compute cell of about 12,500 machines (the Google cluster trace of May 2011, clusterdata-2011-2, CC-BY 4.0): every task's submission time, job, priority, CPU and memory request and run time, and every machine's capacity, failure time and return time, are taken from that trace (ids are renamed). Where and when each task ran in the small cluster is not recorded in the trace: it is computed from those records by the scheduling policy stated below, applied exactly.

The document is divided into chapters titled 'Cell window <NN>'. Each chapter is an independent run of one small cluster over a window of 1200 seconds. Questions refer to a chapter by that title. Times are whole seconds from the start of the window (time 0); the window covers times 0 to 1200.

Each chapter has four sections. 'Machines' lists the cluster's machines with their CPU and memory capacity, the time a machine failed (if it did) and the time it returned (if it did). 'Task log' lists, in time order, every event of every task from time 0 to the window end: submit, start, finish, evict and fail, with the task's job, priority, CPU and memory request, run time, and the machine involved. 'Events' lists machine failures and returns, evictions and restarts, and the longest waits. 'Queue at window end' gives the state after the events at time 1200: the tasks still running (with their remaining run time) and the tasks still waiting, in queue order.

Tasks. A task belongs to a job and has a priority from 0 to 11 (higher is more important). Priorities 9 to 11 are the *production* tier and priorities 0 and 1 the *free* tier. Each task requests an amount of CPU and of memory, which it holds on one machine while it runs, and has a run time: once started it runs for exactly that many seconds and then finishes, unless it is stopped first. The run time is given on every row of the task log, starting with the task's submit row.

Scheduling policy. Every window starts with all machines empty and no task waiting. At every time at which something happens, in this order:

1. **Finishes.** Tasks whose run time is over finish and free their CPU and memory.

2. **Failures.** Machines that fail at this time go down. Each task running on such a machine stops (a *fail* event) and goes back to the queue.

3. **Returns.** Machines that return at this time come back, empty, with their capacity.

4. **Submissions.** Tasks submitted at this time join the queue.

5. **Scheduling round.** The waiting tasks are taken one at a time in *queue order*: higher priority first, then earlier submission time, then lower task id. Each task, in turn:

a. **First fit.** The task starts on the first machine, in machine id order, that is up and whose free CPU and free memory are both at least the task's requests (subject to the reservation rule in c).

b. **Preemption.** If it fits nowhere, no task holds a reservation yet in this round, and it is a production task, it may evict free-tier tasks from one machine. It looks at the up machines in id order. On each it considers the free-tier tasks running there, lowest priority first, then the most recently started first (then the higher task id first), and evicts them one by one in that order until it fits; it takes the first machine where this works, evicting only as many as needed there, and starts on it. If no machine works, nothing is evicted. An evicted task goes back to the queue at once (it may start again later in the same round); it loses its progress and will run its full run time again.

c. **Reservation and backfilling.** The first task in the round that can start neither by a. nor by b. gets a *reservation*: the earliest time at which it could start, computed from the run times of the tasks now running (assuming no new submissions, failures or returns), on the first machine, in id order, where that earliest time is reached. Every later task in the round may still start by first fit, but on the reserved machine only if it will finish no later than the reserved time, or if its requests fit in what the reserved machine will still have free at the reserved time after the reserved task's requests (and after other such starts on it in the same round). On the other machines it starts as usual. Later tasks do not preempt. A task that would not fit on any up machine even if that machine were empty gets no reservation and simply keeps waiting; the next task that cannot start may then get the reservation.

Tasks stopped by an eviction or a failure keep their original submission time (and so their place in queue order) and start again from the beginning. A task never moves while it runs.

Order of rows. Task log rows with the same time follow the steps above: finishes (by task id), failures (by machine, then task id), submissions (by task id), then the scheduling round's starts and evictions in the order the round makes them (each eviction just before the start that caused it).

Terms used in the Events section and in questions. A task's *wait* is the time from its submission to its first start. A task is *restarted* when it starts again after an eviction or a failure. The number of *evictions* counts evict events (one per evicted task). 'Running on a machine at time t' means running after all the events at time t.

The cluster and the workload. Each window has 8 machines, ids M01 to M08. Each is modelled on a real machine of the cell (one of them is a machine that failed during the window in the trace) and has 1/8 of that machine's recorded CPU and memory capacity. A failed machine returns at the time the trace records it coming back, if that is within 6 hours; otherwise it does not return. The tasks are a sample of the real tasks submitted in the cell during the window, chosen by this fixed rule: jobs submitted in the window are taken in a random order, first up to 6 production jobs, then free-tier jobs until the offered load reaches 54%, then other jobs until it reaches 90% (a job that would push it above 95% is skipped). A job contributes its first 4 tasks (16 for a free-tier job) that finished normally in the trace and fit on at least one machine of the window. Offered load = the larger, over CPU and memory, of the sum of request x run time inside the window over all tasks, divided by the cluster's total capacity x the window length.

After the window. For questions about what happens after time 1200: no new task is submitted and no machine fails after the window; a machine that is down returns at its listed return time (if any); the policy continues unchanged from the state given in 'Queue at window end'.

Units. Times and run times in seconds. CPU and memory in units of 1/100,000 of the largest machine in the original cell (the trace's normalised values, rounded to whole units).

Chapters in this document (itops_dev): Cell window 08, Cell window 17, Cell window 27, Cell window 30.

Cell window 08

Times 0 to 1200. The cluster has 8 machines. 85 tasks of 43 jobs are submitted during the window.

Cell window 08: Machines

machine	CPU	memory	fails at	returns at
M01	6250	6244	-	-
M02	6250	3116	424	never
M03	6250	3116	-	-
M04	6250	6244	-	-
M05	6250	6244	-	-
M06	6250	9363	-	-
M07	6250	3116	-	-
M08	6250	3116	-	-

Cell window 08: Task log

time	task	job	priority	CPU	memory	run time	event	machine
27	T001	J01	8	625	16	35	submit	-
27	T001	J01	8	625	16	35	start	M01
43	T002	J02	8	937	466	67	submit	-
43	T002	J02	8	937	466	67	start	M01
46	T003	J03	4	625	197	66	submit	-
46	T003	J03	4	625	197	66	start	M01
62	T001	J01	8	625	16	35	finish	M01
104	T004	J04	0	6250	1244	88	submit	-
104	T004	J04	0	6250	1244	88	start	M02
110	T002	J02	8	937	466	67	finish	M01
112	T003	J03	4	625	197	66	finish	M01
126	T005	J05	4	625	155	30	submit	-
126	T006	J06	4	625	311	329	submit	-
126	T007	J07	4	625	311	29	submit	-
126	T008	J08	4	3125	1125	29	submit	-
126	T005	J05	4	625	155	30	start	M01
126	T006	J06	4	625	311	329	start	M01
126	T007	J07	4	625	311	29	start	M01
126	T008	J08	4	3125	1125	29	start	M01
155	T007	J07	4	625	311	29	finish	M01
155	T008	J08	4	3125	1125	29	finish	M01
156	T005	J05	4	625	155	30	finish	M01
171	T009	J09	0	625	219	261	submit	-
171	T010	J09	0	625	219	260	submit	-
171	T011	J09	0	625	219	258	submit	-
171	T012	J09	0	625	219	258	submit	-
171	T013	J09	0	625	219	259	submit	-
171	T014	J09	0	625	219	258	submit	-
171	T015	J09	0	625	219	257	submit	-
171	T016	J09	0	625	219	257	submit	-
171	T017	J09	0	625	219	257	submit	-
171	T018	J09	0	625	219	257	submit	-
171	T019	J09	0	625	219	260	submit	-
171	T020	J09	0	625	219	257	submit	-
171	T021	J09	0	625	219	260	submit	-
171	T022	J09	0	625	219	259	submit	-
171	T023	J09	0	625	219	259	submit	-
171	T024	J09	0	625	219	261	submit	-

time	task	job	priority	CPU	memory	run time	event	machine
171	T009	J09	0	625	219	261	start	M01
171	T010	J09	0	625	219	260	start	M01
171	T011	J09	0	625	219	258	start	M01
171	T012	J09	0	625	219	258	start	M01
171	T013	J09	0	625	219	259	start	M01
171	T014	J09	0	625	219	258	start	M01
171	T015	J09	0	625	219	257	start	M01
171	T016	J09	0	625	219	257	start	M01
171	T017	J09	0	625	219	257	start	M01
171	T018	J09	0	625	219	257	start	M03
171	T019	J09	0	625	219	260	start	M03
171	T020	J09	0	625	219	257	start	M03
171	T021	J09	0	625	219	260	start	M03
171	T022	J09	0	625	219	259	start	M03
171	T023	J09	0	625	219	259	start	M03
171	T024	J09	0	625	219	261	start	M03
192	T004	J04	0	6250	1244	88	finish	M02
239	T025	J10	4	1250	610	779	submit	-
239	T026	J10	4	1250	610	737	submit	-
239	T027	J10	4	1250	610	650	submit	-
239	T028	J10	4	1250	610	640	submit	-
239	T025	J10	4	1250	610	779	start	M02
239	T026	J10	4	1250	610	737	start	M02
239	T027	J10	4	1250	610	650	start	M02
239	T028	J10	4	1250	610	640	start	M02
403	T029	J11	9	6250	3180	330	submit	-
403	T029	J11	9	6250	3180	330	start	M04
424	T025	J10	4	1250	610	779	fail	M02
424	T026	J10	4	1250	610	737	fail	M02
424	T027	J10	4	1250	610	650	fail	M02
424	T028	J10	4	1250	610	640	fail	M02
424	T025	J10	4	1250	610	779	start	M03
424	T026	J10	4	1250	610	737	start	M05
424	T027	J10	4	1250	610	650	start	M05
424	T028	J10	4	1250	610	640	start	M05
428	T015	J09	0	625	219	257	finish	M01
428	T016	J09	0	625	219	257	finish	M01
428	T017	J09	0	625	219	257	finish	M01
428	T018	J09	0	625	219	257	finish	M03
428	T020	J09	0	625	219	257	finish	M03
429	T011	J09	0	625	219	258	finish	M01
429	T012	J09	0	625	219	258	finish	M01
429	T014	J09	0	625	219	258	finish	M01
430	T013	J09	0	625	219	259	finish	M01
430	T022	J09	0	625	219	259	finish	M03
430	T023	J09	0	625	219	259	finish	M03
431	T010	J09	0	625	219	260	finish	M01
431	T019	J09	0	625	219	260	finish	M03
431	T021	J09	0	625	219	260	finish	M03
432	T009	J09	0	625	219	261	finish	M01
432	T024	J09	0	625	219	261	finish	M03
455	T006	J06	4	625	311	329	finish	M01
549	T030	J12	1	4376	777	3126	submit	-
549	T030	J12	1	4376	777	3126	start	M01
583	T031	J13	9	62	2	26	submit	-
583	T031	J13	9	62	2	26	start	M01
585	T032	J14	0	6250	1244	89	submit	-

time	task	job	priority	CPU	memory	run time	event	machine
585	T032	J14	0	6250	1244	89	start	M06
597	T033	J15	0	3125	311	2278	submit	-
597	T034	J15	0	3125	311	2498	submit	-
597	T035	J15	0	3125	311	2133	submit	-
597	T036	J15	0	3125	311	2452	submit	-
597	T037	J15	0	3125	311	2133	submit	-
597	T038	J15	0	3125	311	2098	submit	-
597	T039	J15	0	3125	311	2134	submit	-
597	T040	J15	0	3125	311	2227	submit	-
597	T041	J15	0	3125	311	2116	submit	-
597	T042	J15	0	3125	311	2427	submit	-
597	T043	J15	0	3125	311	2133	submit	-
597	T044	J15	0	3125	311	2094	submit	-
597	T045	J15	0	3125	311	2097	submit	-
597	T046	J15	0	3125	311	2102	submit	-
597	T047	J15	0	3125	311	2094	submit	-
597	T048	J15	0	3125	311	2115	submit	-
597	T033	J15	0	3125	311	2278	start	M03
597	T034	J15	0	3125	311	2498	start	M07
597	T035	J15	0	3125	311	2133	start	M07
597	T036	J15	0	3125	311	2452	start	M08
597	T037	J15	0	3125	311	2133	start	M08
609	T031	J13	9	62	2	26	finish	M01
643	T049	J16	8	3125	622	50	submit	-
643	T050	J17	4	3125	1125	35	submit	-
648	T051	J18	9	6250	1590	92	submit	-
648	T030	J12	1	4376	777	3126	evict	M01
648	T051	J18	9	6250	1590	92	start	M01
659	T052	J19	8	625	466	34	submit	-
659	T052	J19	8	625	466	34	start	M03
674	T032	J14	0	6250	1244	89	finish	M06
674	T049	J16	8	3125	622	50	start	M06
674	T050	J17	4	3125	1125	35	start	M06
693	T052	J19	8	625	466	34	finish	M03
704	T053	J20	0	6250	466	44	submit	-
709	T050	J17	4	3125	1125	35	finish	M06
714	T054	J21	4	625	155	27	submit	-
714	T055	J22	4	1250	398	32	submit	-
714	T056	J23	4	3125	1125	43	submit	-
714	T054	J21	4	625	155	27	start	M03
714	T055	J22	4	1250	398	32	start	M03
714	T056	J23	4	3125	1125	43	start	M06
724	T049	J16	8	3125	622	50	finish	M06
724	T038	J15	0	3125	311	2098	start	M06
733	T029	J11	9	6250	3180	330	finish	M04
733	T030	J12	1	4376	777	3126	start	M04
740	T051	J18	9	6250	1590	92	finish	M01
740	T039	J15	0	3125	311	2134	start	M01
740	T040	J15	0	3125	311	2227	start	M01
741	T054	J21	4	625	155	27	finish	M03
746	T055	J22	4	1250	398	32	finish	M03
757	T056	J23	4	3125	1125	43	finish	M06
757	T041	J15	0	3125	311	2116	start	M06
825	T057	J24	9	4688	6360	579	submit	-
825	T041	J15	0	3125	311	2116	evict	M06
825	T038	J15	0	3125	311	2098	evict	M06
825	T057	J24	9	4688	6360	579	start	M06

time	task	job	priority	CPU	memory	run time	event	machine
1040	T058	J25	6	625	187	152	submit	-
1040	T059	J26	6	6250	622	123	submit	-
1040	T060	J27	6	625	280	47	submit	-
1040	T061	J28	6	625	187	142	submit	-
1040	T062	J29	6	6250	622	134	submit	-
1040	T063	J30	6	625	280	57	submit	-
1040	T064	J31	6	6250	622	182	submit	-
1040	T065	J32	6	625	280	107	submit	-
1040	T066	J33	4	62	398	32	submit	-
1040	T067	J34	4	625	155	32	submit	-
1040	T068	J35	8	6250	3180	861	submit	-
1040	T069	J36	4	1250	466	253	submit	-
1040	T058	J25	6	625	187	152	start	M03
1040	T060	J27	6	625	280	47	start	M03
1040	T061	J28	6	625	187	142	start	M03
1040	T063	J30	6	625	280	57	start	M04
1040	T065	J32	6	625	280	107	start	M04
1040	T066	J33	4	62	398	32	start	M04
1040	T067	J34	4	625	155	32	start	M05
1040	T069	J36	4	1250	466	253	start	M06
1042	T070	J37	4	2493	2704	860	submit	-
1042	T071	J37	4	2493	2704	858	submit	-
1042	T072	J37	4	2493	2704	857	submit	-
1042	T073	J37	4	2493	2704	859	submit	-
1042	T074	J38	4	2933	2704	464	submit	-
1042	T075	J38	4	2933	2704	466	submit	-
1042	T076	J38	4	2933	2704	468	submit	-
1042	T077	J38	4	2933	2704	550	submit	-
1042	T078	J39	4	1250	466	224	submit	-
1042	T079	J39	4	1250	466	236	submit	-
1042	T080	J39	4	1250	466	240	submit	-
1042	T081	J39	4	1250	466	235	submit	-
1060	T082	J40	8	3125	388	30	submit	-
1064	T028	J10	4	1250	610	640	finish	M05
1064	T082	J40	8	3125	388	30	start	M05
1066	T083	J41	0	6250	1244	57	submit	-
1072	T066	J33	4	62	398	32	finish	M04
1072	T067	J34	4	625	155	32	finish	M05
1074	T027	J10	4	1250	610	650	finish	M05
1087	T060	J27	6	625	280	47	finish	M03
1094	T082	J40	8	3125	388	30	finish	M05
1097	T063	J30	6	625	280	57	finish	M04
1108	T084	J42	6	625	187	20	submit	-
1108	T084	J42	6	625	187	20	start	M03
1128	T084	J42	6	625	187	20	finish	M03
1147	T065	J32	6	625	280	107	finish	M04
1147	T078	J39	4	1250	466	224	start	M04
1160	T085	J43	6	625	187	27	submit	-
1160	T085	J43	6	625	187	27	start	M03
1161	T026	J10	4	1250	610	737	finish	M05
1161	T068	J35	8	6250	3180	861	start	M05
1182	T061	J28	6	625	187	142	finish	M03
1187	T085	J43	6	625	187	27	finish	M03
1187	T079	J39	4	1250	466	236	start	M03
1192	T058	J25	6	625	187	152	finish	M03

Cell window 08: Events

Time 424: machine M02 failed; tasks T025, T026, T027, T028 stopped and went back to the queue.

Time 424: task T025 restarted on machine M03.

Time 424: task T026 restarted on machine M05.

Time 424: task T027 restarted on machine M05.

Time 424: task T028 restarted on machine M05.

Time 648: task T051 (priority 9) evicted task T030 (priority 1) from machine M01.

Time 733: task T030 restarted on machine M04.

Time 825: task T057 (priority 9) evicted task T038 (priority 0) from machine M06.

Time 825: task T057 (priority 9) evicted task T041 (priority 0) from machine M06.

Longest waits (submission to first start, tasks started by the window end): task T041 waited 160 s (submitted at 597, first started at 757); task T079 waited 145 s (submitted at 1042, first started at 1187); task T039 waited 143 s (submitted at 597, first started at 740); task T040 waited 143 s (submitted at 597, first started at 740); task T038 waited 127 s (submitted at 597, first started at 724).

Cell window 08: Queue at window end

State after the events at time 1200.

Tasks running at time 1200 (remaining run time = seconds still to run after time 1200):

task	job	priority	CPU	memory	machine	started	remaining
T039	J15	0	3125	311	M01	740	1674
T040	J15	0	3125	311	M01	740	1767
T025	J10	4	1250	610	M03	424	3
T033	J15	0	3125	311	M03	597	1675
T079	J39	4	1250	466	M03	1187	223
T030	J12	1	4376	777	M04	733	2659
T078	J39	4	1250	466	M04	1147	171
T068	J35	8	6250	3180	M05	1161	822
T057	J24	9	4688	6360	M06	825	204
T069	J36	4	1250	466	M06	1040	93
T034	J15	0	3125	311	M07	597	1895
T035	J15	0	3125	311	M07	597	1530
T036	J15	0	3125	311	M08	597	1849
T037	J15	0	3125	311	M08	597	1530

Tasks waiting at time 1200, in queue order (each will run its full run time once started):

task	job	priority	CPU	memory	submitted	run time
T059	J26	6	6250	622	1040	123
T062	J29	6	6250	622	1040	134
T064	J31	6	6250	622	1040	182
T070	J37	4	2493	2704	1042	860
T071	J37	4	2493	2704	1042	858
T072	J37	4	2493	2704	1042	857
T073	J37	4	2493	2704	1042	859
T074	J38	4	2933	2704	1042	464
T075	J38	4	2933	2704	1042	466
T076	J38	4	2933	2704	1042	468
T077	J38	4	2933	2704	1042	550
T080	J39	4	1250	466	1042	240
T081	J39	4	1250	466	1042	235

task	job	priority	CPU	memory	submitted	run time
T038	J15	0	3125	311	597	2098
T041	J15	0	3125	311	597	2116
T042	J15	0	3125	311	597	2427
T043	J15	0	3125	311	597	2133
T044	J15	0	3125	311	597	2094
T045	J15	0	3125	311	597	2097
T046	J15	0	3125	311	597	2102
T047	J15	0	3125	311	597	2094
T048	J15	0	3125	311	597	2115
T053	J20	0	6250	466	704	44
T083	J41	0	6250	1244	1066	57

Machine M02 is down at time 1200; it does not return.

Cell window 17

Times 0 to 1200. The cluster has 8 machines. 75 tasks of 36 jobs are submitted during the window.

Cell window 17: Machines

machine	CPU	memory	fails at	returns at
M01	6250	3116	-	-
M02	6250	6244	-	-
M03	6250	6244	-	-
M04	6250	9363	-	-
M05	6250	3116	312	1312
M06	6250	6244	-	-
M07	6250	3116	-	-
M08	3125	3123	-	-

Cell window 17: Task log

time	task	job	priority	CPU	memory	run time	event	machine
15	T001	J01	4	625	155	35	submit	-
15	T001	J01	4	625	155	35	start	M01
50	T001	J01	4	625	155	35	finish	M01
62	T002	J02	0	125	1553	5443	submit	-
62	T003	J02	0	125	1553	5499	submit	-
62	T004	J02	0	125	1553	5442	submit	-
62	T005	J02	0	125	1553	5461	submit	-
62	T006	J02	0	125	1553	5482	submit	-
62	T007	J02	0	125	1553	5469	submit	-
62	T008	J02	0	125	1553	5496	submit	-
62	T009	J02	0	125	1553	5423	submit	-
62	T010	J02	0	125	1553	5474	submit	-
62	T011	J02	0	125	1553	5516	submit	-
62	T012	J02	0	125	1553	5474	submit	-
62	T013	J02	0	125	1553	5498	submit	-
62	T014	J02	0	125	1553	5430	submit	-
62	T015	J02	0	125	1553	5442	submit	-
62	T016	J02	0	125	1553	5472	submit	-
62	T017	J02	0	125	1553	5430	submit	-
62	T002	J02	0	125	1553	5443	start	M01
62	T003	J02	0	125	1553	5499	start	M01
62	T004	J02	0	125	1553	5442	start	M02
62	T005	J02	0	125	1553	5461	start	M02
62	T006	J02	0	125	1553	5482	start	M02
62	T007	J02	0	125	1553	5469	start	M02
62	T008	J02	0	125	1553	5496	start	M03
62	T009	J02	0	125	1553	5423	start	M03
62	T010	J02	0	125	1553	5474	start	M03
62	T011	J02	0	125	1553	5516	start	M03
62	T012	J02	0	125	1553	5474	start	M04
62	T013	J02	0	125	1553	5498	start	M04
62	T014	J02	0	125	1553	5430	start	M04
62	T015	J02	0	125	1553	5442	start	M04
62	T016	J02	0	125	1553	5472	start	M04
62	T017	J02	0	125	1553	5430	start	M04
65	T018	J03	9	4688	6360	544	submit	-
65	T017	J02	0	125	1553	5430	evict	M04
65	T016	J02	0	125	1553	5472	evict	M04

time	task	job	priority	CPU	memory	run time	event	machine
65	T015	J02	0	125	1553	5442	evict	M04
65	T014	J02	0	125	1553	5430	evict	M04
65	T013	J02	0	125	1553	5498	evict	M04
65	T018	J03	9	4688	6360	544	start	M04
65	T013	J02	0	125	1553	5498	start	M05
65	T014	J02	0	125	1553	5430	start	M05
65	T015	J02	0	125	1553	5442	start	M06
65	T016	J02	0	125	1553	5472	start	M06
65	T017	J02	0	125	1553	5430	start	M06
84	T019	J04	6	625	280	56	submit	-
84	T020	J05	6	625	187	130	submit	-
84	T019	J04	6	625	280	56	start	M04
84	T020	J05	6	625	187	130	start	M04
140	T019	J04	6	625	280	56	finish	M04
150	T021	J06	6	6250	622	344	submit	-
150	T022	J07	6	625	280	36	submit	-
150	T023	J08	8	625	311	448	submit	-
150	T023	J08	8	625	311	448	start	M04
150	T021	J06	6	6250	622	344	start	M07
150	T022	J07	6	625	280	36	start	M06
186	T022	J07	6	625	280	36	finish	M06
214	T020	J05	6	625	187	130	finish	M04
229	T024	J09	6	625	280	323	submit	-
229	T024	J09	6	625	280	323	start	M04
304	T025	J10	9	6250	3180	200	submit	-
304	T007	J02	0	125	1553	5469	evict	M02
304	T006	J02	0	125	1553	5482	evict	M02
304	T005	J02	0	125	1553	5461	evict	M02
304	T004	J02	0	125	1553	5442	evict	M02
304	T025	J10	9	6250	3180	200	start	M02
304	T004	J02	0	125	1553	5442	start	M06
304	T005	J02	0	125	1553	5461	start	M08
304	T006	J02	0	125	1553	5482	start	M08
312	T013	J02	0	125	1553	5498	fail	M05
312	T014	J02	0	125	1553	5430	fail	M05
318	T026	J11	4	625	155	36	submit	-
436	T027	J12	1	625	16	735	submit	-
436	T027	J12	1	625	16	735	start	M03
494	T021	J06	6	6250	622	344	finish	M07
494	T026	J11	4	625	155	36	start	M07
494	T007	J02	0	125	1553	5469	start	M07
499	T028	J13	8	625	1590	295	submit	-
504	T025	J10	9	6250	3180	200	finish	M02
504	T028	J13	8	625	1590	295	start	M02
504	T013	J02	0	125	1553	5498	start	M02
504	T014	J02	0	125	1553	5430	start	M02
527	T029	J14	6	1250	613	317	submit	-
527	T030	J14	6	1250	613	289	submit	-
527	T031	J14	6	1250	613	311	submit	-
527	T032	J14	6	1250	613	312	submit	-
527	T029	J14	6	1250	613	317	start	M02
527	T030	J14	6	1250	613	289	start	M02
527	T031	J14	6	1250	613	311	start	M07
527	T032	J14	6	1250	613	312	start	M07
530	T026	J11	4	625	155	36	finish	M07
547	T033	J15	0	6250	1244	40	submit	-
552	T024	J09	6	625	280	323	finish	M04

time	task	job	priority	CPU	memory	run time	event	machine
568	T034	J16	6	625	280	105	submit	-
568	T034	J16	6	625	280	105	start	M02
598	T023	J08	8	625	311	448	finish	M04
604	T035	J17	9	6250	233	100	submit	-
604	T003	J02	0	125	1553	5499	evict	M01
604	T002	J02	0	125	1553	5443	evict	M01
604	T035	J17	9	6250	233	100	start	M01
609	T018	J03	9	4688	6360	544	finish	M04
609	T002	J02	0	125	1553	5443	start	M04
609	T003	J02	0	125	1553	5499	start	M04
611	T036	J18	9	6250	1590	70	submit	-
611	T011	J02	0	125	1553	5516	evict	M03
611	T010	J02	0	125	1553	5474	evict	M03
611	T009	J02	0	125	1553	5423	evict	M03
611	T008	J02	0	125	1553	5496	evict	M03
611	T027	J12	1	625	16	735	evict	M03
611	T036	J18	9	6250	1590	70	start	M03
611	T027	J12	1	625	16	735	start	M02
611	T008	J02	0	125	1553	5496	start	M04
611	T009	J02	0	125	1553	5423	start	M04
611	T010	J02	0	125	1553	5474	start	M04
641	T037	J19	0	1250	622	50	submit	-
641	T038	J19	0	1250	622	46	submit	-
641	T039	J19	0	1250	622	47	submit	-
641	T040	J19	0	1250	622	50	submit	-
641	T041	J19	0	1250	622	47	submit	-
641	T042	J19	0	1250	622	50	submit	-
641	T043	J19	0	1250	622	48	submit	-
654	T044	J20	4	3125	932	453	submit	-
654	T045	J20	4	3125	932	507	submit	-
654	T046	J20	4	3125	932	435	submit	-
654	T047	J20	4	3125	932	482	submit	-
673	T034	J16	6	625	280	105	finish	M02
679	T048	J21	6	625	280	247	submit	-
679	T048	J21	6	625	280	247	start	M02
681	T036	J18	9	6250	1590	70	finish	M03
681	T044	J20	4	3125	932	453	start	M03
681	T045	J20	4	3125	932	507	start	M03
693	T049	J22	4	1874	1424	173	submit	-
693	T050	J22	4	1874	1424	191	submit	-
693	T051	J22	4	1874	1424	169	submit	-
693	T052	J22	4	1874	1424	172	submit	-
704	T035	J17	9	6250	233	100	finish	M01
704	T046	J20	4	3125	932	435	start	M01
704	T047	J20	4	3125	932	482	start	M01
710	T053	J23	6	1250	595	189	submit	-
710	T054	J23	6	1250	595	187	submit	-
710	T055	J23	6	1250	595	186	submit	-
710	T056	J23	6	1250	595	186	submit	-
750	T057	J24	6	6250	622	117	submit	-
778	T058	J25	4	1250	861	2813	submit	-
778	T059	J25	4	1250	861	2882	submit	-
778	T060	J25	4	1250	861	2315	submit	-
778	T061	J25	4	1250	861	1976	submit	-
786	T062	J26	9	4688	6360	531	submit	-
786	T010	J02	0	125	1553	5474	evict	M04
786	T009	J02	0	125	1553	5423	evict	M04

time	task	job	priority	CPU	memory	run time	event	machine
786	T008	J02	0	125	1553	5496	evict	M04
786	T003	J02	0	125	1553	5499	evict	M04
786	T002	J02	0	125	1553	5443	evict	M04
786	T062	J26	9	4688	6360	531	start	M04
786	T053	J23	6	1250	595	189	start	M04
799	T028	J13	8	625	1590	295	finish	M02
799	T054	J23	6	1250	595	187	start	M02
804	T063	J27	6	625	187	22	submit	-
804	T063	J27	6	625	187	22	start	M02
816	T030	J14	6	1250	613	289	finish	M02
816	T055	J23	6	1250	595	186	start	M02
826	T063	J27	6	625	187	22	finish	M02
838	T031	J14	6	1250	613	311	finish	M07
838	T056	J23	6	1250	595	186	start	M07
839	T032	J14	6	1250	613	312	finish	M07
839	T058	J25	4	1250	861	2813	start	M07
844	T029	J14	6	1250	613	317	finish	M02
844	T049	J22	4	1874	1424	173	start	M02
880	T064	J28	6	625	280	360	submit	-
925	T065	J29	4	625	311	26	submit	-
926	T048	J21	6	625	280	247	finish	M02
926	T064	J28	6	625	280	360	start	M02
937	T066	J30	6	625	280	190	submit	-
975	T053	J23	6	1250	595	189	finish	M04
975	T066	J30	6	625	280	190	start	M04
975	T065	J29	4	625	311	26	start	M04
986	T054	J23	6	1250	595	187	finish	M02
986	T037	J19	0	1250	622	50	start	M02
999	T067	J31	8	5627	4773	782	submit	-
1001	T065	J29	4	625	311	26	finish	M04
1002	T055	J23	6	1250	595	186	finish	M02
1002	T038	J19	0	1250	622	46	start	M02
1017	T049	J22	4	1874	1424	173	finish	M02
1017	T050	J22	4	1874	1424	191	start	M02
1024	T056	J23	6	1250	595	186	finish	M07
1024	T039	J19	0	1250	622	47	start	M07
1028	T068	J32	6	625	280	49	submit	-
1028	T068	J32	6	625	280	49	start	M04
1036	T037	J19	0	1250	622	50	finish	M02
1036	T040	J19	0	1250	622	50	start	M02
1045	T069	J33	8	1874	3180	289	submit	-
1048	T038	J19	0	1250	622	46	finish	M02
1048	T041	J19	0	1250	622	47	start	M02
1071	T039	J19	0	1250	622	47	finish	M07
1071	T042	J19	0	1250	622	50	start	M07
1077	T068	J32	6	625	280	49	finish	M04
1086	T040	J19	0	1250	622	50	finish	M02
1086	T043	J19	0	1250	622	48	start	M02
1095	T041	J19	0	1250	622	47	finish	M02
1105	T070	J34	8	625	398	1010	submit	-
1105	T070	J34	8	625	398	1010	start	M02
1121	T042	J19	0	1250	622	50	finish	M07
1134	T043	J19	0	1250	622	48	finish	M02
1134	T044	J20	4	3125	932	453	finish	M03
1134	T059	J25	4	1250	861	2882	start	M02
1139	T046	J20	4	3125	932	435	finish	M01
1139	T051	J22	4	1874	1424	169	start	M01

time	task	job	priority	CPU	memory	run time	event	machine
1157	T071	J35	8	5627	3497	808	submit	-
1158	T072	J36	4	1874	3180	763	submit	-
1158	T073	J36	4	1874	3180	761	submit	-
1158	T074	J36	4	1874	3180	759	submit	-
1158	T075	J36	4	1874	3180	775	submit	-
1165	T066	J30	6	625	280	190	finish	M04
1165	T060	J25	4	1250	861	2315	start	M04
1186	T047	J20	4	3125	932	482	finish	M01
1186	T052	J22	4	1874	1424	172	start	M01
1188	T045	J20	4	3125	932	507	finish	M03
1188	T067	J31	8	5627	4773	782	start	M03

Cell window 17: Events

Time 65: task T018 (priority 9) evicted task T013 (priority 0) from machine M04.

Time 65: task T018 (priority 9) evicted task T014 (priority 0) from machine M04.

Time 65: task T018 (priority 9) evicted task T015 (priority 0) from machine M04.

Time 65: task T018 (priority 9) evicted task T016 (priority 0) from machine M04.

Time 65: task T018 (priority 9) evicted task T017 (priority 0) from machine M04.

Time 65: task T013 restarted on machine M05.

Time 65: task T014 restarted on machine M05.

Time 65: task T015 restarted on machine M06.

Time 65: task T016 restarted on machine M06.

Time 65: task T017 restarted on machine M06.

Time 304: task T025 (priority 9) evicted task T004 (priority 0) from machine M02.

Time 304: task T025 (priority 9) evicted task T005 (priority 0) from machine M02.

Time 304: task T025 (priority 9) evicted task T006 (priority 0) from machine M02.

Time 304: task T025 (priority 9) evicted task T007 (priority 0) from machine M02.

Time 304: task T004 restarted on machine M06.

Time 304: task T005 restarted on machine M08.

Time 304: task T006 restarted on machine M08.

Time 312: machine M05 failed; tasks T013, T014 stopped and went back to the queue.

Time 494: task T007 restarted on machine M07.

Time 504: task T013 restarted on machine M02.

Time 504: task T014 restarted on machine M02.

Time 604: task T035 (priority 9) evicted task T002 (priority 0) from machine M01.

Time 604: task T035 (priority 9) evicted task T003 (priority 0) from machine M01.

Time 609: task T002 restarted on machine M04.

Time 609: task T003 restarted on machine M04.

Time 611: task T036 (priority 9) evicted task T008 (priority 0) from machine M03.

Time 611: task T036 (priority 9) evicted task T009 (priority 0) from machine M03.

Time 611: task T036 (priority 9) evicted task T010 (priority 0) from machine M03.

Time 611: task T036 (priority 9) evicted task T011 (priority 0) from machine M03.

Time 611: task T036 (priority 9) evicted task T027 (priority 1) from machine M03.

Time 611: task T008 restarted on machine M04.

Time 611: task T009 restarted on machine M04.

Time 611: task T010 restarted on machine M04.

Time 611: task T027 restarted on machine M02.

Time 786: task T062 (priority 9) evicted task T002 (priority 0) from machine M04.

Time 786: task T062 (priority 9) evicted task T003 (priority 0) from machine M04.

Time 786: task T062 (priority 9) evicted task T008 (priority 0) from machine M04.

Time 786: task T062 (priority 9) evicted task T009 (priority 0) from machine M04.

Time 786: task T062 (priority 9) evicted task T010 (priority 0) from machine M04.

Longest waits (submission to first start, tasks started by the window end): task T052 waited 493 s (submitted at 693, first started at 1186); task T051 waited 446 s (submitted at 693, first started at 1139); task T043 waited 445 s (submitted at 641, first started at 1086); task T042 waited 430 s (submitted at 641, first started at 1071); task T041 waited 407 s (submitted at 641, first started at 1048).

Cell window 17: Queue at window end

State after the events at time 1200.

Tasks running at time 1200 (remaining run time = seconds still to run after time 1200):

task	job	priority	CPU	memory	machine	started	remaining
T051	J22	4	1874	1424	M01	1139	108
T052	J22	4	1874	1424	M01	1186	158
T013	J02	0	125	1553	M02	504	4802
T014	J02	0	125	1553	M02	504	4734
T027	J12	1	625	16	M02	611	146
T050	J22	4	1874	1424	M02	1017	8
T059	J25	4	1250	861	M02	1134	2816
T064	J28	6	625	280	M02	926	86
T070	J34	8	625	398	M02	1105	915
T067	J31	8	5627	4773	M03	1188	770
T012	J02	0	125	1553	M04	62	4336
T060	J25	4	1250	861	M04	1165	2280
T062	J26	9	4688	6360	M04	786	117
T004	J02	0	125	1553	M06	304	4546
T015	J02	0	125	1553	M06	65	4307
T016	J02	0	125	1553	M06	65	4337
T017	J02	0	125	1553	M06	65	4295
T007	J02	0	125	1553	M07	494	4763
T058	J25	4	1250	861	M07	839	2452
T005	J02	0	125	1553	M08	304	4565
T006	J02	0	125	1553	M08	304	4586

Tasks waiting at time 1200, in queue order (each will run its full run time once started):

task	job	priority	CPU	memory	submitted	run time
T069	J33	8	1874	3180	1045	289
T071	J35	8	5627	3497	1157	808
T057	J24	6	6250	622	750	117
T061	J25	4	1250	861	778	1976
T072	J36	4	1874	3180	1158	763

task	job	priority	CPU	memory	submitted	run time
T073	J36	4	1874	3180	1158	761
T074	J36	4	1874	3180	1158	759
T075	J36	4	1874	3180	1158	775
T002	J02	0	125	1553	62	5443
T003	J02	0	125	1553	62	5499
T008	J02	0	125	1553	62	5496
T009	J02	0	125	1553	62	5423
T010	J02	0	125	1553	62	5474
T011	J02	0	125	1553	62	5516
T033	J15	0	6250	1244	547	40

Machine M05 is down at time 1200; it returns at time 1312.

Cell window 27

Times 0 to 1200. The cluster has 8 machines. 111 tasks of 51 jobs are submitted during the window.

Cell window 27: Machines

machine	CPU	memory	fails at	returns at
M01	6250	6244	-	-
M02	6250	6244	-	-
M03	6250	3116	-	-
M04	12500	12500	437	549
M05	12500	12500	-	-
M06	6250	3116	-	-
M07	6250	6244	-	-
M08	6250	6244	-	-

Cell window 27: Task log

time	task	job	priority	CPU	memory	run time	event	machine
38	T001	J01	0	3125	466	266	submit	-
38	T001	J01	0	3125	466	266	start	M01
98	T002	J02	0	6250	466	38	submit	-
98	T002	J02	0	6250	466	38	start	M02
99	T003	J03	1	1250	1590	374	submit	-
99	T003	J03	1	1250	1590	374	start	M01
100	T004	J04	1	2499	311	252	submit	-
100	T004	J04	1	2499	311	252	start	M03
110	T005	J05	9	6250	1590	79	submit	-
110	T005	J05	9	6250	1590	79	start	M04
114	T006	J06	6	625	280	113	submit	-
114	T007	J07	4	625	311	25	submit	-
114	T006	J06	6	625	280	113	start	M01
114	T007	J07	4	625	311	25	start	M01
136	T002	J02	0	6250	466	38	finish	M02
139	T007	J07	4	625	311	25	finish	M01
142	T008	J08	1	3748	466	197	submit	-
142	T008	J08	1	3748	466	197	start	M02
145	T009	J09	6	645	197	72	submit	-
145	T010	J10	0	1524	271	174	submit	-
145	T011	J10	0	1524	271	175	submit	-
145	T012	J10	0	1524	271	175	submit	-
145	T013	J10	0	1524	271	177	submit	-
145	T014	J10	0	1524	271	175	submit	-
145	T015	J10	0	1524	271	177	submit	-
145	T016	J10	0	1524	271	173	submit	-
145	T017	J10	0	1524	271	175	submit	-
145	T018	J10	0	1524	271	173	submit	-
145	T019	J10	0	1524	271	176	submit	-
145	T020	J10	0	1524	271	175	submit	-
145	T021	J10	0	1524	271	174	submit	-
145	T022	J10	0	1524	271	176	submit	-
145	T023	J10	0	1524	271	173	submit	-
145	T024	J10	0	1524	271	174	submit	-
145	T025	J10	0	1524	271	176	submit	-
145	T009	J09	6	645	197	72	start	M01
145	T010	J10	0	1524	271	174	start	M02
145	T011	J10	0	1524	271	175	start	M03

time	task	job	priority	CPU	memory	run time	event	machine
145	T012	J10	0	1524	271	175	start	M03
145	T013	J10	0	1524	271	177	start	M04
145	T014	J10	0	1524	271	175	start	M04
145	T015	J10	0	1524	271	177	start	M04
145	T016	J10	0	1524	271	173	start	M04
145	T017	J10	0	1524	271	175	start	M05
145	T018	J10	0	1524	271	173	start	M05
145	T019	J10	0	1524	271	176	start	M05
145	T020	J10	0	1524	271	175	start	M05
145	T021	J10	0	1524	271	174	start	M05
145	T022	J10	0	1524	271	176	start	M05
145	T023	J10	0	1524	271	173	start	M05
145	T024	J10	0	1524	271	174	start	M05
145	T025	J10	0	1524	271	176	start	M06
186	T026	J11	1	1250	777	1139	submit	-
186	T026	J11	1	1250	777	1139	start	M06
189	T005	J05	9	6250	1590	79	finish	M04
217	T009	J09	6	645	197	72	finish	M01
219	T027	J12	9	4688	6360	487	submit	-
219	T027	J12	9	4688	6360	487	start	M04
220	T028	J13	1	625	622	783	submit	-
220	T028	J13	1	625	622	783	start	M01
227	T006	J06	6	625	280	113	finish	M01
238	T029	J14	0	1688	295	1070	submit	-
238	T029	J14	0	1688	295	1070	start	M04
304	T001	J01	0	3125	466	266	finish	M01
318	T016	J10	0	1524	271	173	finish	M04
318	T018	J10	0	1524	271	173	finish	M05
318	T023	J10	0	1524	271	173	finish	M05
319	T010	J10	0	1524	271	174	finish	M02
319	T021	J10	0	1524	271	174	finish	M05
319	T024	J10	0	1524	271	174	finish	M05
320	T011	J10	0	1524	271	175	finish	M03
320	T012	J10	0	1524	271	175	finish	M03
320	T014	J10	0	1524	271	175	finish	M04
320	T017	J10	0	1524	271	175	finish	M05
320	T020	J10	0	1524	271	175	finish	M05
321	T019	J10	0	1524	271	176	finish	M05
321	T022	J10	0	1524	271	176	finish	M05
321	T025	J10	0	1524	271	176	finish	M06
322	T013	J10	0	1524	271	177	finish	M04
322	T015	J10	0	1524	271	177	finish	M04
337	T030	J15	0	6250	1244	33	submit	-
337	T030	J15	0	6250	1244	33	start	M05
339	T008	J08	1	3748	466	197	finish	M02
352	T004	J04	1	2499	311	252	finish	M03
353	T031	J16	1	625	16	739	submit	-
353	T031	J16	1	625	16	739	start	M01
360	T032	J17	1	4376	777	634	submit	-
360	T032	J17	1	4376	777	634	start	M02
370	T030	J15	0	6250	1244	33	finish	M05
399	T033	J18	0	1250	16	10	submit	-
399	T033	J18	0	1250	16	10	start	M01
402	T034	J19	9	6250	233	110	submit	-
402	T034	J19	9	6250	233	110	start	M03
405	T035	J20	1	3125	777	576	submit	-
405	T035	J20	1	3125	777	576	start	M04

time	task	job	priority	CPU	memory	run time	event	machine
407	T036	J21	8	3125	388	20	submit	-
407	T037	J22	4	6873	1910	2936	submit	-
407	T038	J22	4	6873	1910	2954	submit	-
407	T039	J22	4	6873	1910	3065	submit	-
407	T040	J22	4	6873	1910	2937	submit	-
407	T041	J23	0	1494	311	573	submit	-
407	T036	J21	8	3125	388	20	start	M05
407	T037	J22	4	6873	1910	2936	start	M05
407	T041	J23	0	1494	311	573	start	M01
408	T042	J24	0	625	311	360	submit	-
408	T043	J24	0	625	311	346	submit	-
408	T044	J24	0	625	311	346	submit	-
408	T045	J24	0	625	311	368	submit	-
408	T046	J24	0	625	311	409	submit	-
408	T047	J24	0	625	311	336	submit	-
408	T048	J24	0	625	311	348	submit	-
408	T049	J24	0	625	311	573	submit	-
408	T050	J24	0	625	311	348	submit	-
408	T051	J24	0	625	311	337	submit	-
408	T052	J24	0	625	311	418	submit	-
408	T053	J24	0	625	311	375	submit	-
408	T054	J24	0	625	311	334	submit	-
408	T055	J24	0	625	311	359	submit	-
408	T056	J24	0	625	311	437	submit	-
408	T057	J24	0	625	311	378	submit	-
408	T042	J24	0	625	311	360	start	M01
408	T043	J24	0	625	311	346	start	M02
408	T044	J24	0	625	311	346	start	M02
408	T045	J24	0	625	311	368	start	M04
408	T046	J24	0	625	311	409	start	M05
408	T047	J24	0	625	311	336	start	M05
408	T048	J24	0	625	311	348	start	M05
408	T049	J24	0	625	311	573	start	M05
408	T050	J24	0	625	311	348	start	M06
408	T051	J24	0	625	311	337	start	M06
408	T052	J24	0	625	311	418	start	M06
408	T053	J24	0	625	311	375	start	M06
408	T054	J24	0	625	311	334	start	M06
408	T055	J24	0	625	311	359	start	M06
408	T056	J24	0	625	311	437	start	M06
408	T057	J24	0	625	311	378	start	M07
409	T033	J18	0	1250	16	10	finish	M01
425	T058	J25	4	3125	1125	26	submit	-
425	T058	J25	4	3125	1125	26	start	M07
427	T036	J21	8	3125	388	20	finish	M05
437	T027	J12	9	4688	6360	487	fail	M04
437	T029	J14	0	1688	295	1070	fail	M04
437	T035	J20	1	3125	777	576	fail	M04
437	T045	J24	0	625	311	368	fail	M04
437	T049	J24	0	625	311	573	evict	M05
437	T048	J24	0	625	311	348	evict	M05
437	T047	J24	0	625	311	336	evict	M05
437	T027	J12	9	4688	6360	487	start	M05
437	T035	J20	1	3125	777	576	start	M08
437	T029	J14	0	1688	295	1070	start	M07
437	T045	J24	0	625	311	368	start	M01
437	T047	J24	0	625	311	336	start	M01

time	task	job	priority	CPU	memory	run time	event	machine
437	T048	J24	0	625	311	348	start	M07
437	T049	J24	0	625	311	573	start	M08
445	T059	J26	1	1250	777	1121	submit	-
445	T059	J26	1	1250	777	1121	start	M08
451	T058	J25	4	3125	1125	26	finish	M07
473	T003	J03	1	1250	1590	374	finish	M01
488	T060	J27	0	1688	295	1065	submit	-
488	T060	J27	0	1688	295	1065	start	M07
512	T034	J19	9	6250	233	110	finish	M03
549	T038	J22	4	6873	1910	2954	start	M04
593	T061	J28	4	3125	1125	28	submit	-
593	T061	J28	4	3125	1125	28	start	M03
621	T061	J28	4	3125	1125	28	finish	M03
673	T062	J29	8	3125	3180	147	submit	-
673	T063	J30	8	625	155	213	submit	-
673	T062	J29	8	3125	3180	147	start	M04
673	T063	J30	8	625	155	213	start	M01
699	T064	J31	0	6250	466	39	submit	-
699	T064	J31	0	6250	466	39	start	M03
708	T065	J32	9	6250	1590	70	submit	-
708	T044	J24	0	625	311	346	evict	M02
708	T043	J24	0	625	311	346	evict	M02
708	T032	J17	1	4376	777	634	evict	M02
708	T065	J32	9	6250	1590	70	start	M02
708	T043	J24	0	625	311	346	start	M01
708	T044	J24	0	625	311	346	start	M04
738	T064	J31	0	6250	466	39	finish	M03
738	T032	J17	1	4376	777	634	start	M03
742	T054	J24	0	625	311	334	finish	M06
743	T066	J33	8	937	1553	64	submit	-
743	T066	J33	8	937	1553	64	start	M03
745	T051	J24	0	625	311	337	finish	M06
756	T050	J24	0	625	311	348	finish	M06
767	T055	J24	0	625	311	359	finish	M06
768	T042	J24	0	625	311	360	finish	M01
773	T047	J24	0	625	311	336	finish	M01
778	T065	J32	9	6250	1590	70	finish	M02
782	T067	J34	8	1250	2863	2183	submit	-
782	T067	J34	8	1250	2863	2183	start	M01
783	T053	J24	0	625	311	375	finish	M06
785	T048	J24	0	625	311	348	finish	M07
786	T057	J24	0	625	311	378	finish	M07
805	T045	J24	0	625	311	368	finish	M01
807	T066	J33	8	937	1553	64	finish	M03
817	T046	J24	0	625	311	409	finish	M05
817	T068	J35	9	4688	6360	485	submit	-
817	T069	J36	0	6250	1244	58	submit	-
817	T069	J36	0	6250	1244	58	start	M02
820	T062	J29	8	3125	3180	147	finish	M04
820	T068	J35	9	4688	6360	485	start	M04
826	T052	J24	0	625	311	418	finish	M06
828	T070	J37	4	62	398	54	submit	-
828	T070	J37	4	62	398	54	start	M01
838	T071	J38	1	625	311	53	submit	-
838	T071	J38	1	625	311	53	start	M01
845	T056	J24	0	625	311	437	finish	M06
875	T069	J36	0	6250	1244	58	finish	M02

time	task	job	priority	CPU	memory	run time	event	machine
876	T072	J39	0	6250	1590	4040	submit	-
876	T072	J39	0	6250	1590	4040	start	M02
882	T070	J37	4	62	398	54	finish	M01
886	T063	J30	8	625	155	213	finish	M01
891	T071	J38	1	625	311	53	finish	M01
909	T073	J40	4	2234	3122	269	submit	-
909	T074	J40	4	2234	3122	269	submit	-
909	T075	J40	4	2234	3122	264	submit	-
909	T076	J40	4	2234	3122	260	submit	-
909	T073	J40	4	2234	3122	269	start	M07
924	T027	J12	9	4688	6360	487	finish	M05
924	T074	J40	4	2234	3122	269	start	M05
924	T075	J40	4	2234	3122	264	start	M05
938	T077	J41	0	3125	466	273	submit	-
938	T077	J41	0	3125	466	273	start	M06
969	T078	J42	1	6250	466	619	submit	-
970	T079	J43	0	2594	197	617	submit	-
970	T080	J43	0	2594	197	617	submit	-
970	T081	J43	0	2594	197	616	submit	-
970	T082	J43	0	2594	197	610	submit	-
970	T083	J43	0	2594	197	616	submit	-
970	T084	J43	0	2594	197	608	submit	-
970	T085	J43	0	2594	197	617	submit	-
970	T086	J43	0	2594	197	609	submit	-
970	T087	J43	0	2594	197	609	submit	-
970	T088	J43	0	2594	197	616	submit	-
980	T041	J23	0	1494	311	573	finish	M01
980	T079	J43	0	2594	197	617	start	M01
1003	T028	J13	1	625	622	783	finish	M01
1004	T089	J44	1	625	622	37	submit	-
1004	T089	J44	1	625	622	37	start	M01
1010	T049	J24	0	625	311	573	finish	M08
1013	T035	J20	1	3125	777	576	finish	M08
1013	T076	J40	4	2234	3122	260	start	M08
1013	T080	J43	0	2594	197	617	start	M08
1015	T090	J45	4	3125	777	47	submit	-
1041	T089	J44	1	625	622	37	finish	M01
1054	T043	J24	0	625	311	346	finish	M01
1054	T044	J24	0	625	311	346	finish	M04
1092	T031	J16	1	625	16	739	finish	M01
1094	T091	J46	1	2499	311	245	submit	-
1120	T092	J47	1	3748	466	203	submit	-
1124	T093	J48	0	1520	271	171	submit	-
1124	T094	J48	0	1520	271	173	submit	-
1124	T095	J48	0	1520	271	169	submit	-
1124	T096	J48	0	1520	271	170	submit	-
1124	T097	J48	0	1520	271	186	submit	-
1124	T098	J48	0	1520	271	173	submit	-
1124	T099	J48	0	1520	271	168	submit	-
1124	T100	J48	0	1520	271	188	submit	-
1124	T101	J48	0	1520	271	190	submit	-
1124	T102	J48	0	1520	271	169	submit	-
1124	T103	J48	0	1520	271	169	submit	-
1124	T104	J48	0	1520	271	182	submit	-
1124	T105	J48	0	1520	271	188	submit	-
1124	T106	J48	0	1520	271	177	submit	-
1124	T107	J48	0	1520	271	180	submit	-

time	task	job	priority	CPU	memory	run time	event	machine
1124	T108	J48	0	1520	271	173	submit	-
1124	T093	J48	0	1520	271	171	start	M01
1124	T094	J48	0	1520	271	173	start	M03
1124	T095	J48	0	1520	271	169	start	M06
1162	T109	J49	8	937	466	67	submit	-
1162	T110	J50	8	937	466	57	submit	-
1162	T109	J49	8	937	466	67	start	M04
1162	T110	J50	8	937	466	57	start	M05
1165	T111	J51	4	1250	197	55	submit	-
1178	T073	J40	4	2234	3122	269	finish	M07
1178	T111	J51	4	1250	197	55	start	M07
1178	T096	J48	0	1520	271	170	start	M07
1188	T075	J40	4	2234	3122	264	finish	M05
1188	T097	J48	0	1520	271	186	start	M05
1193	T074	J40	4	2234	3122	269	finish	M05
1193	T090	J45	4	3125	777	47	start	M05

Cell window 27: Events

Time 437: machine M04 failed; tasks T027, T029, T035, T045 stopped and went back to the queue.

Time 437: task T027 (priority 9) evicted task T047 (priority 0) from machine M05.

Time 437: task T027 (priority 9) evicted task T048 (priority 0) from machine M05.

Time 437: task T027 (priority 9) evicted task T049 (priority 0) from machine M05.

Time 437: task T027 restarted on machine M05.

Time 437: task T029 restarted on machine M07.

Time 437: task T035 restarted on machine M08.

Time 437: task T045 restarted on machine M01.

Time 437: task T047 restarted on machine M01.

Time 437: task T048 restarted on machine M07.

Time 437: task T049 restarted on machine M08.

Time 549: machine M04 returned.

Time 708: task T065 (priority 9) evicted task T032 (priority 1) from machine M02.

Time 708: task T065 (priority 9) evicted task T043 (priority 0) from machine M02.

Time 708: task T065 (priority 9) evicted task T044 (priority 0) from machine M02.

Time 708: task T043 restarted on machine M01.

Time 708: task T044 restarted on machine M04.

Time 738: task T032 restarted on machine M03.

Longest waits (submission to first start, tasks started by the window end): task T090 waited 178 s (submitted at 1015, first started at 1193); task T038 waited 142 s (submitted at 407, first started at 549); task T076 waited 104 s (submitted at 909, first started at 1013); task T097 waited 64 s (submitted at 1124, first started at 1188); task T096 waited 54 s (submitted at 1124, first started at 1178).

Cell window 27: Queue at window end

State after the events at time 1200.

Tasks running at time 1200 (remaining run time = seconds still to run after time 1200):

task	job	priority	CPU	memory	machine	started	remaining
T067	J34	8	1250	2863	M01	782	1765
T079	J43	0	2594	197	M01	980	397
T093	J48	0	1520	271	M01	1124	95
T072	J39	0	6250	1590	M02	876	3716
T032	J17	1	4376	777	M03	738	172
T094	J48	0	1520	271	M03	1124	97
T038	J22	4	6873	1910	M04	549	2303
T068	J35	9	4688	6360	M04	820	105
T109	J49	8	937	466	M04	1162	29
T037	J22	4	6873	1910	M05	407	2143
T090	J45	4	3125	777	M05	1193	40
T097	J48	0	1520	271	M05	1188	174
T110	J50	8	937	466	M05	1162	19
T026	J11	1	1250	777	M06	186	125
T077	J41	0	3125	466	M06	938	11
T095	J48	0	1520	271	M06	1124	93
T029	J14	0	1688	295	M07	437	307
T060	J27	0	1688	295	M07	488	353
T096	J48	0	1520	271	M07	1178	148
T111	J51	4	1250	197	M07	1178	33
T059	J26	1	1250	777	M08	445	366
T076	J40	4	2234	3122	M08	1013	73
T080	J43	0	2594	197	M08	1013	430

Tasks waiting at time 1200, in queue order (each will run its full run time once started):

task	job	priority	CPU	memory	submitted	run time
T039	J22	4	6873	1910	407	3065
T040	J22	4	6873	1910	407	2937
T078	J42	1	6250	466	969	619
T091	J46	1	2499	311	1094	245
T092	J47	1	3748	466	1120	203
T081	J43	0	2594	197	970	616
T082	J43	0	2594	197	970	610
T083	J43	0	2594	197	970	616
T084	J43	0	2594	197	970	608
T085	J43	0	2594	197	970	617
T086	J43	0	2594	197	970	609
T087	J43	0	2594	197	970	609
T088	J43	0	2594	197	970	616
T098	J48	0	1520	271	1124	173
T099	J48	0	1520	271	1124	168
T100	J48	0	1520	271	1124	188
T101	J48	0	1520	271	1124	190
T102	J48	0	1520	271	1124	169
T103	J48	0	1520	271	1124	169
T104	J48	0	1520	271	1124	182
T105	J48	0	1520	271	1124	188
T106	J48	0	1520	271	1124	177
T107	J48	0	1520	271	1124	180
T108	J48	0	1520	271	1124	173

All machines are up at time 1200.

Cell window 30

Times 0 to 1200. The cluster has 8 machines. 155 tasks of 44 jobs are submitted during the window.

Cell window 30: Machines

machine	CPU	memory	fails at	returns at
M01	6250	6244	-	-
M02	6250	3116	-	-
M03	12500	12500	-	-
M04	6250	6244	-	-
M05	6250	3116	-	-
M06	6250	6244	-	-
M07	6250	6244	484	878
M08	6250	9363	-	-

Cell window 30: Task log

time	task	job	priority	CPU	memory	run time	event	machine
6	T001	J01	8	1250	777	70	submit	-
6	T001	J01	8	1250	777	70	start	M01
43	T002	J02	0	6250	466	34	submit	-
43	T002	J02	0	6250	466	34	start	M02
44	T003	J03	1	1250	1590	386	submit	-
44	T003	J03	1	1250	1590	386	start	M01
45	T004	J04	9	6250	233	103	submit	-
45	T004	J04	9	6250	233	103	start	M03
48	T005	J05	9	6250	1590	75	submit	-
48	T005	J05	9	6250	1590	75	start	M03
76	T001	J01	8	1250	777	70	finish	M01
77	T002	J02	0	6250	466	34	finish	M02
90	T006	J06	1	1250	466	151	submit	-
90	T006	J06	1	1250	466	151	start	M01
92	T007	J07	0	625	242	149	submit	-
92	T008	J07	0	625	242	149	submit	-
92	T009	J07	0	625	242	148	submit	-
92	T010	J07	0	625	242	147	submit	-
92	T011	J07	0	625	242	149	submit	-
92	T012	J07	0	625	242	147	submit	-
92	T013	J07	0	625	242	148	submit	-
92	T014	J07	0	625	242	148	submit	-
92	T015	J07	0	625	242	148	submit	-
92	T016	J07	0	625	242	148	submit	-
92	T017	J07	0	625	242	149	submit	-
92	T018	J07	0	625	242	147	submit	-
92	T019	J07	0	625	242	148	submit	-
92	T020	J07	0	625	242	148	submit	-
92	T021	J07	0	625	242	149	submit	-
92	T022	J07	0	625	242	148	submit	-
92	T007	J07	0	625	242	149	start	M01
92	T008	J07	0	625	242	149	start	M01
92	T009	J07	0	625	242	148	start	M01
92	T010	J07	0	625	242	147	start	M01
92	T011	J07	0	625	242	149	start	M01
92	T012	J07	0	625	242	147	start	M01
92	T013	J07	0	625	242	148	start	M02
92	T014	J07	0	625	242	148	start	M02

time	task	job	priority	CPU	memory	run time	event	machine
92	T015	J07	0	625	242	148	start	M02
92	T016	J07	0	625	242	148	start	M02
92	T017	J07	0	625	242	149	start	M02
92	T018	J07	0	625	242	147	start	M02
92	T019	J07	0	625	242	148	start	M02
92	T020	J07	0	625	242	148	start	M02
92	T021	J07	0	625	242	149	start	M02
92	T022	J07	0	625	242	148	start	M02
106	T023	J08	1	3125	622	90	submit	-
106	T023	J08	1	3125	622	90	start	M04
123	T005	J05	9	6250	1590	75	finish	M03
136	T024	J09	0	1688	292	1066	submit	-
136	T024	J09	0	1688	292	1066	start	M03
148	T004	J04	9	6250	233	103	finish	M03
163	T025	J10	9	4688	6360	482	submit	-
163	T025	J10	9	4688	6360	482	start	M03
196	T023	J08	1	3125	622	90	finish	M04
239	T010	J07	0	625	242	147	finish	M01
239	T012	J07	0	625	242	147	finish	M01
239	T018	J07	0	625	242	147	finish	M02
240	T009	J07	0	625	242	148	finish	M01
240	T013	J07	0	625	242	148	finish	M02
240	T014	J07	0	625	242	148	finish	M02
240	T015	J07	0	625	242	148	finish	M02
240	T016	J07	0	625	242	148	finish	M02
240	T019	J07	0	625	242	148	finish	M02
240	T020	J07	0	625	242	148	finish	M02
240	T022	J07	0	625	242	148	finish	M02
241	T006	J06	1	1250	466	151	finish	M01
241	T007	J07	0	625	242	149	finish	M01
241	T008	J07	0	625	242	149	finish	M01
241	T011	J07	0	625	242	149	finish	M01
241	T017	J07	0	625	242	149	finish	M02
241	T021	J07	0	625	242	149	finish	M02
257	T026	J11	8	3125	3180	146	submit	-
257	T027	J12	8	625	155	201	submit	-
257	T026	J11	8	3125	3180	146	start	M01
257	T027	J12	8	625	155	201	start	M01
263	T028	J13	0	937	274	100	submit	-
263	T029	J13	0	937	274	159	submit	-
263	T030	J13	0	937	274	93	submit	-
263	T031	J13	0	937	274	92	submit	-
263	T032	J13	0	937	274	133	submit	-
263	T033	J13	0	937	274	87	submit	-
263	T034	J13	0	937	274	100	submit	-
263	T035	J13	0	937	274	110	submit	-
263	T036	J13	0	937	274	111	submit	-
263	T037	J13	0	937	274	93	submit	-
263	T038	J13	0	937	274	144	submit	-
263	T039	J13	0	937	274	102	submit	-
263	T040	J13	0	937	274	121	submit	-
263	T041	J13	0	937	274	128	submit	-
263	T042	J13	0	937	274	100	submit	-
263	T043	J13	0	937	274	100	submit	-
263	T028	J13	0	937	274	100	start	M01
263	T029	J13	0	937	274	159	start	M02
263	T030	J13	0	937	274	93	start	M02

time	task	job	priority	CPU	memory	run time	event	machine
263	T031	J13	0	937	274	92	start	M02
263	T032	J13	0	937	274	133	start	M02
263	T033	J13	0	937	274	87	start	M02
263	T034	J13	0	937	274	100	start	M02
263	T035	J13	0	937	274	110	start	M03
263	T036	J13	0	937	274	111	start	M03
263	T037	J13	0	937	274	93	start	M03
263	T038	J13	0	937	274	144	start	M03
263	T039	J13	0	937	274	102	start	M03
263	T040	J13	0	937	274	121	start	M03
263	T041	J13	0	937	274	128	start	M04
263	T042	J13	0	937	274	100	start	M04
263	T043	J13	0	937	274	100	start	M04
284	T044	J14	0	6250	1244	36	submit	-
284	T044	J14	0	6250	1244	36	start	M05
285	T045	J15	0	6250	1244	59	submit	-
285	T045	J15	0	6250	1244	59	start	M06
320	T044	J14	0	6250	1244	36	finish	M05
322	T046	J16	1	1250	777	1135	submit	-
322	T046	J16	1	1250	777	1135	start	M04
339	T047	J17	4	4999	1590	2271	submit	-
339	T047	J17	4	4999	1590	2271	start	M05
344	T045	J15	0	6250	1244	59	finish	M06
350	T033	J13	0	937	274	87	finish	M02
355	T031	J13	0	937	274	92	finish	M02
356	T030	J13	0	937	274	93	finish	M02
356	T037	J13	0	937	274	93	finish	M03
362	T048	J18	1	6250	466	662	submit	-
362	T048	J18	1	6250	466	662	start	M06
363	T028	J13	0	937	274	100	finish	M01
363	T034	J13	0	937	274	100	finish	M02
363	T042	J13	0	937	274	100	finish	M04
363	T043	J13	0	937	274	100	finish	M04
365	T039	J13	0	937	274	102	finish	M03
373	T035	J13	0	937	274	110	finish	M03
374	T036	J13	0	937	274	111	finish	M03
384	T040	J13	0	937	274	121	finish	M03
391	T041	J13	0	937	274	128	finish	M04
396	T032	J13	0	937	274	133	finish	M02
403	T026	J11	8	3125	3180	146	finish	M01
407	T038	J13	0	937	274	144	finish	M03
422	T029	J13	0	937	274	159	finish	M02
430	T003	J03	1	1250	1590	386	finish	M01
458	T027	J12	8	625	155	201	finish	M01
474	T049	J19	1	1353	777	1093	submit	-
474	T049	J19	1	1353	777	1093	start	M01
478	T050	J20	0	937	272	89	submit	-
478	T051	J20	0	937	272	83	submit	-
478	T052	J20	0	937	272	93	submit	-
478	T053	J20	0	937	272	79	submit	-
478	T054	J20	0	937	272	181	submit	-
478	T055	J20	0	937	272	102	submit	-
478	T056	J20	0	937	272	79	submit	-
478	T057	J20	0	937	272	83	submit	-
478	T058	J20	0	937	272	80	submit	-
478	T059	J20	0	937	272	98	submit	-
478	T060	J20	0	937	272	92	submit	-

time	task	job	priority	CPU	memory	run time	event	machine
478	T061	J20	0	937	272	83	submit	-
478	T062	J20	0	937	272	80	submit	-
478	T063	J20	0	937	272	80	submit	-
478	T064	J20	0	937	272	82	submit	-
478	T065	J20	0	937	272	92	submit	-
478	T050	J20	0	937	272	89	start	M01
478	T051	J20	0	937	272	83	start	M01
478	T052	J20	0	937	272	93	start	M01
478	T053	J20	0	937	272	79	start	M01
478	T054	J20	0	937	272	181	start	M01
478	T055	J20	0	937	272	102	start	M02
478	T056	J20	0	937	272	79	start	M02
478	T057	J20	0	937	272	83	start	M02
478	T058	J20	0	937	272	80	start	M02
478	T059	J20	0	937	272	98	start	M02
478	T060	J20	0	937	272	92	start	M02
478	T061	J20	0	937	272	83	start	M03
478	T062	J20	0	937	272	80	start	M03
478	T063	J20	0	937	272	80	start	M03
478	T064	J20	0	937	272	82	start	M03
478	T065	J20	0	937	272	92	start	M03
520	T066	J21	8	937	466	81	submit	-
520	T066	J21	8	937	466	81	start	M03
557	T053	J20	0	937	272	79	finish	M01
557	T056	J20	0	937	272	79	finish	M02
558	T058	J20	0	937	272	80	finish	M02
558	T062	J20	0	937	272	80	finish	M03
558	T063	J20	0	937	272	80	finish	M03
560	T064	J20	0	937	272	82	finish	M03
561	T051	J20	0	937	272	83	finish	M01
561	T057	J20	0	937	272	83	finish	M02
561	T061	J20	0	937	272	83	finish	M03
567	T050	J20	0	937	272	89	finish	M01
570	T060	J20	0	937	272	92	finish	M02
570	T065	J20	0	937	272	92	finish	M03
571	T052	J20	0	937	272	93	finish	M01
576	T059	J20	0	937	272	98	finish	M02
580	T055	J20	0	937	272	102	finish	M02
584	T067	J22	1	3125	3180	192	submit	-
584	T067	J22	1	3125	3180	192	start	M01
601	T066	J21	8	937	466	81	finish	M03
642	T068	J23	0	6250	466	38	submit	-
642	T068	J23	0	6250	466	38	start	M02
645	T025	J10	9	4688	6360	482	finish	M03
647	T069	J24	1	1572	3180	118	submit	-
647	T070	J25	9	6250	1590	69	submit	-
647	T070	J25	9	6250	1590	69	start	M03
647	T069	J24	1	1572	3180	118	start	M03
648	T071	J26	0	1250	967	95	submit	-
648	T072	J26	0	1250	967	92	submit	-
648	T073	J26	0	1250	967	94	submit	-
648	T074	J26	0	1250	967	92	submit	-
648	T075	J26	0	1250	967	92	submit	-
648	T076	J26	0	1250	967	95	submit	-
648	T077	J26	0	1250	967	102	submit	-
648	T078	J26	0	1250	967	103	submit	-
648	T079	J26	0	1250	967	113	submit	-

time	task	job	priority	CPU	memory	run time	event	machine
648	T080	J26	0	1250	967	101	submit	-
648	T081	J26	0	1250	967	103	submit	-
648	T082	J26	0	1250	967	96	submit	-
648	T083	J26	0	1250	967	113	submit	-
648	T084	J26	0	1250	967	116	submit	-
648	T085	J26	0	1250	967	101	submit	-
648	T086	J26	0	1250	967	112	submit	-
648	T071	J26	0	1250	967	95	start	M03
648	T072	J26	0	1250	967	92	start	M03
648	T073	J26	0	1250	967	94	start	M04
648	T074	J26	0	1250	967	92	start	M04
648	T075	J26	0	1250	967	92	start	M04
648	T076	J26	0	1250	967	95	start	M04
648	T077	J26	0	1250	967	102	start	M05
648	T078	J26	0	1250	967	103	start	M08
648	T079	J26	0	1250	967	113	start	M08
648	T080	J26	0	1250	967	101	start	M08
648	T081	J26	0	1250	967	103	start	M08
648	T082	J26	0	1250	967	96	start	M08
650	T087	J27	0	937	966	80	submit	-
650	T088	J27	0	937	966	81	submit	-
650	T089	J27	0	937	966	93	submit	-
650	T090	J27	0	937	966	110	submit	-
650	T091	J27	0	937	966	79	submit	-
650	T092	J27	0	937	966	80	submit	-
650	T093	J27	0	937	966	80	submit	-
650	T094	J27	0	937	966	78	submit	-
650	T095	J27	0	937	966	93	submit	-
650	T096	J27	0	937	966	111	submit	-
650	T097	J27	0	937	966	110	submit	-
650	T098	J27	0	937	966	95	submit	-
650	T099	J27	0	937	966	94	submit	-
650	T100	J27	0	937	966	92	submit	-
650	T101	J27	0	937	966	79	submit	-
650	T102	J27	0	937	966	81	submit	-
659	T054	J20	0	937	272	181	finish	M01
659	T083	J26	0	1250	967	113	start	M01
680	T068	J23	0	6250	466	38	finish	M02
680	T084	J26	0	1250	967	116	start	M02
680	T085	J26	0	1250	967	101	start	M02
680	T086	J26	0	1250	967	112	start	M02
702	T103	J28	4	2756	2704	735	submit	-
702	T104	J28	4	2756	2704	739	submit	-
702	T105	J28	4	2756	2704	747	submit	-
702	T106	J28	4	2756	2704	736	submit	-
702	T107	J29	4	625	208	73	submit	-
702	T108	J29	4	625	208	72	submit	-
702	T109	J29	4	625	208	73	submit	-
702	T110	J29	4	625	208	72	submit	-
702	T107	J29	4	625	208	73	start	M02
716	T070	J25	9	6250	1590	69	finish	M03
716	T103	J28	4	2756	2704	735	start	M03
716	T104	J28	4	2756	2704	739	start	M03
716	T108	J29	4	625	208	72	start	M03
740	T072	J26	0	1250	967	92	finish	M03
740	T074	J26	0	1250	967	92	finish	M04
740	T075	J26	0	1250	967	92	finish	M04

time	task	job	priority	CPU	memory	run time	event	machine
740	T109	J29	4	625	208	73	start	M03
740	T110	J29	4	625	208	72	start	M03
740	T087	J27	0	937	966	80	start	M04
742	T073	J26	0	1250	967	94	finish	M04
742	T105	J28	4	2756	2704	747	start	M04
743	T071	J26	0	1250	967	95	finish	M03
743	T076	J26	0	1250	967	95	finish	M04
743	T088	J27	0	937	966	81	start	M03
743	T089	J27	0	937	966	93	start	M04
744	T082	J26	0	1250	967	96	finish	M08
744	T090	J27	0	937	966	110	start	M08
749	T080	J26	0	1250	967	101	finish	M08
749	T091	J27	0	937	966	79	start	M08
750	T077	J26	0	1250	967	102	finish	M05
750	T092	J27	0	937	966	80	start	M05
751	T078	J26	0	1250	967	103	finish	M08
751	T081	J26	0	1250	967	103	finish	M08
751	T106	J28	4	2756	2704	736	start	M08
761	T079	J26	0	1250	967	113	finish	M08
761	T093	J27	0	937	966	80	start	M08
764	T111	J30	9	4688	6360	496	submit	-
764	T088	J27	0	937	966	81	evict	M03
764	T024	J09	0	1688	292	1066	evict	M03
764	T069	J24	1	1572	3180	118	evict	M03
764	T111	J30	9	4688	6360	496	start	M03
766	T112	J31	4	2725	2704	454	submit	-
766	T113	J31	4	2725	2704	441	submit	-
766	T114	J31	4	2725	2704	440	submit	-
766	T115	J31	4	2725	2704	443	submit	-
772	T083	J26	0	1250	967	113	finish	M01
772	T024	J09	0	1688	292	1066	start	M01
775	T107	J29	4	625	208	73	finish	M02
776	T067	J22	1	3125	3180	192	finish	M01
776	T112	J31	4	2725	2704	454	start	M01
781	T085	J26	0	1250	967	101	finish	M02
788	T108	J29	4	625	208	72	finish	M03
792	T086	J26	0	1250	967	112	finish	M02
796	T084	J26	0	1250	967	116	finish	M02
796	T113	J31	4	2725	2704	441	start	M02
809	T116	J32	4	937	213	48	submit	-
809	T117	J32	4	937	213	48	submit	-
809	T118	J32	4	937	213	45	submit	-
809	T119	J32	4	937	213	49	submit	-
809	T116	J32	4	937	213	48	start	M02
809	T117	J32	4	937	213	48	start	M03
812	T110	J29	4	625	208	72	finish	M03
813	T109	J29	4	625	208	73	finish	M03
813	T118	J32	4	937	213	45	start	M03
820	T087	J27	0	937	966	80	finish	M04
820	T119	J32	4	937	213	49	start	M04
824	T120	J33	1	6250	6360	1244	submit	-
825	T121	J34	1	6250	6360	879	submit	-
826	T122	J35	1	6250	6360	913	submit	-
828	T091	J27	0	937	966	79	finish	M08
830	T092	J27	0	937	966	80	finish	M05
830	T088	J27	0	937	966	81	start	M05
836	T089	J27	0	937	966	93	finish	M04

time	task	job	priority	CPU	memory	run time	event	machine
836	T094	J27	0	937	966	78	start	M04
841	T093	J27	0	937	966	80	finish	M08
854	T090	J27	0	937	966	110	finish	M08
854	T114	J31	4	2725	2704	440	start	M08
857	T116	J32	4	937	213	48	finish	M02
857	T117	J32	4	937	213	48	finish	M03
858	T118	J32	4	937	213	45	finish	M03
869	T119	J32	4	937	213	49	finish	M04
869	T095	J27	0	937	966	93	start	M04
878	T115	J31	4	2725	2704	443	start	M07
878	T069	J24	1	1572	3180	118	start	M07
885	T123	J36	4	3125	777	257	submit	-
885	T124	J36	4	3125	777	255	submit	-
885	T125	J36	4	3125	777	255	submit	-
885	T126	J36	4	3125	777	258	submit	-
885	T127	J37	4	3125	488	223	submit	-
885	T128	J37	4	3125	488	242	submit	-
885	T129	J37	4	3125	488	257	submit	-
885	T130	J37	4	3125	488	228	submit	-
899	T131	J38	4	3125	1125	33	submit	-
911	T088	J27	0	937	966	81	finish	M05
911	T096	J27	0	937	966	111	start	M05
914	T094	J27	0	937	966	78	finish	M04
914	T097	J27	0	937	966	110	start	M04
942	T132	J39	9	6250	233	113	submit	-
942	T048	J18	1	6250	466	662	evict	M06
942	T132	J39	9	6250	233	113	start	M06
962	T095	J27	0	937	966	93	finish	M04
962	T098	J27	0	937	966	95	start	M04
996	T069	J24	1	1572	3180	118	finish	M07
996	T123	J36	4	3125	777	257	start	M07
1022	T096	J27	0	937	966	111	finish	M05
1022	T099	J27	0	937	966	94	start	M05
1024	T097	J27	0	937	966	110	finish	M04
1024	T100	J27	0	937	966	92	start	M04
1039	T133	J40	4	6873	1910	4500	submit	-
1039	T134	J40	4	6873	1910	4504	submit	-
1039	T135	J40	4	6873	1910	4500	submit	-
1039	T136	J40	4	6873	1910	4494	submit	-
1055	T132	J39	9	6250	233	113	finish	M06
1055	T124	J36	4	3125	777	255	start	M06
1055	T125	J36	4	3125	777	255	start	M06
1056	T137	J41	1	6250	466	455	submit	-
1057	T098	J27	0	937	966	95	finish	M04
1057	T101	J27	0	937	966	79	start	M04
1060	T138	J42	0	1953	239	62	submit	-
1060	T139	J42	0	1953	239	229	submit	-
1060	T140	J42	0	1953	239	79	submit	-
1060	T141	J42	0	1953	239	69	submit	-
1060	T142	J42	0	1953	239	61	submit	-
1060	T143	J42	0	1953	239	70	submit	-
1060	T144	J42	0	1953	239	59	submit	-
1060	T145	J42	0	1953	239	68	submit	-
1060	T146	J42	0	1953	239	103	submit	-
1060	T147	J42	0	1953	239	59	submit	-
1060	T148	J42	0	1953	239	63	submit	-
1060	T149	J42	0	1953	239	72	submit	-

time	task	job	priority	CPU	memory	run time	event	machine
1060	T150	J42	0	1953	239	203	submit	-
1060	T151	J42	0	1953	239	68	submit	-
1060	T152	J42	0	1953	239	57	submit	-
1060	T153	J42	0	1953	239	95	submit	-
1060	T138	J42	0	1953	239	62	start	M02
1060	T139	J42	0	1953	239	229	start	M03
1116	T099	J27	0	937	966	94	finish	M05
1116	T100	J27	0	937	966	92	finish	M04
1116	T102	J27	0	937	966	81	start	M04
1122	T138	J42	0	1953	239	62	finish	M02
1122	T140	J42	0	1953	239	79	start	M02
1124	T154	J43	1	6250	6360	3950	submit	-
1125	T155	J44	1	6250	6360	3220	submit	-
1136	T101	J27	0	937	966	79	finish	M04
1197	T102	J27	0	937	966	81	finish	M04
1197	T141	J42	0	1953	239	69	start	M04

Cell window 30: Events

Time 484: machine M07 failed; no task was running on it.

Time 764: task T111 (priority 9) evicted task T024 (priority 0) from machine M03.

Time 764: task T111 (priority 9) evicted task T069 (priority 1) from machine M03.

Time 764: task T111 (priority 9) evicted task T088 (priority 0) from machine M03.

Time 772: task T024 restarted on machine M01.

Time 830: task T088 restarted on machine M05.

Time 878: machine M07 returned.

Time 878: task T069 restarted on machine M07.

Time 942: task T132 (priority 9) evicted task T048 (priority 1) from machine M06.

Longest waits (submission to first start, tasks started by the window end): task T102 waited 466 s (submitted at 650, first started at 1116); task T101 waited 407 s (submitted at 650, first started at 1057); task T100 waited 374 s (submitted at 650, first started at 1024); task T099 waited 372 s (submitted at 650, first started at 1022); task T098 waited 312 s (submitted at 650, first started at 962).

Cell window 30: Queue at window end

State after the events at time 1200.

Tasks running at time 1200 (remaining run time = seconds still to run after time 1200):

task	job	priority	CPU	memory	machine	started	remaining
T024	J09	0	1688	292	M01	772	638
T049	J19	1	1353	777	M01	474	367
T112	J31	4	2725	2704	M01	776	30
T113	J31	4	2725	2704	M02	796	37
T140	J42	0	1953	239	M02	1122	1
T103	J28	4	2756	2704	M03	716	251
T104	J28	4	2756	2704	M03	716	255
T111	J30	9	4688	6360	M03	764	60
T139	J42	0	1953	239	M03	1060	89
T046	J16	1	1250	777	M04	322	257
T105	J28	4	2756	2704	M04	742	289
T141	J42	0	1953	239	M04	1197	66
T047	J17	4	4999	1590	M05	339	1410

task	job	priority	CPU	memory	machine	started	remaining
T124	J36	4	3125	777	M06	1055	110
T125	J36	4	3125	777	M06	1055	110
T115	J31	4	2725	2704	M07	878	121
T123	J36	4	3125	777	M07	996	53
T106	J28	4	2756	2704	M08	751	287
T114	J31	4	2725	2704	M08	854	94

Tasks waiting at time 1200, in queue order (each will run its full run time once started):

task	job	priority	CPU	memory	submitted	run time
T126	J36	4	3125	777	885	258
T127	J37	4	3125	488	885	223
T128	J37	4	3125	488	885	242
T129	J37	4	3125	488	885	257
T130	J37	4	3125	488	885	228
T131	J38	4	3125	1125	899	33
T133	J40	4	6873	1910	1039	4500
T134	J40	4	6873	1910	1039	4504
T135	J40	4	6873	1910	1039	4500
T136	J40	4	6873	1910	1039	4494
T048	J18	1	6250	466	362	662
T120	J33	1	6250	6360	824	1244
T121	J34	1	6250	6360	825	879
T122	J35	1	6250	6360	826	913
T137	J41	1	6250	466	1056	455
T154	J43	1	6250	6360	1124	3950
T155	J44	1	6250	6360	1125	3220
T142	J42	0	1953	239	1060	61
T143	J42	0	1953	239	1060	70
T144	J42	0	1953	239	1060	59
T145	J42	0	1953	239	1060	68
T146	J42	0	1953	239	1060	103
T147	J42	0	1953	239	1060	59
T148	J42	0	1953	239	1060	63
T149	J42	0	1953	239	1060	72
T150	J42	0	1953	239	1060	203
T151	J42	0	1953	239	1060	68
T152	J42	0	1953	239	1060	57
T153	J42	0	1953	239	1060	95

All machines are up at time 1200.